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Palaeomagnetism of the Lisbon Volcanics

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Summary

The natural remanent magnetism of the Eocene Lisbon Volcanics has been determined using 39 samples from 12 lavas. The virtual geomagnetic pole (207° E, 69° N, with $dp = 7{\cdot}7°$ and $dm = 13{\cdot}2°$) is significantly different from the mean Eocene geomagnetic pole for Europe. This result is consistent with previous independent suggestions of the creation of the Bay of Biscay by anticlockwise rotation of the Iberian peninsula, and implies therefore that the rotation was in part Post-Eocene. Alternatively, the Eocene geomagnetic field during the extrusion of the Lisbon Volcanics differed significantly from that recorded by other European Eocene rocks.

Introduction

The geometrical fit, using a least squares criterion, of the 500 fathom submarine contour in the North Atlantic (Bullard *et al.* 1965) as shown in Fig. 1, provides the same suggestion made earlier by Du Toit (1937) and Carey (1955): the Iberian peninsula is envisaged as having rotated about 39° anticlockwise to result in the opening of the Biscay Spenochasm (Carey 1958).

Particularly when originating from more than one source, such an hypothesis obviously warrants testing by any appropriate means. Given fulfilment of several conditions, the palaeomagnetic method is ideally suited to such a study. The principle involved is to compare a palaeomagnetic declination in the hypothetically rotated region with the palaeomagnetic declination resulting from studies of rocks of the same general age in adjacent stable regions. Kawai *et al.* (1961), Irving & Opdyke (1965), Watkins (1964, 1965), De Boer (1965), and others, have used this principle in attempting resolution of megatectonic problems.

The method is ideally applied in regions of shallow palaeomagnetic inclination since accurate definition of the horizontal component diminishes with increasing inclination, becoming indeterminable at the geomagnetic pole. It is also desirable for the mean natural remanent magnetism (NRM) used in testing the hypothesis to result from sampling bodies spanning at least 10 000 years in age (Runcorn 1959, Hide 1967) in order that short period variations may be cancelled to the axial dipole condition: sampling from a restricted time interval could conceivably result in the definition of an inclined dipole or secular variation feature, rendering the results of comparisons with other data less meaningful.

It is the purpose of this communication to present the results of a palaeomagnetic survey of the Lisbon Volcanics. In addition to providing the first palaeomagnetic data from Portugal, the study serves as a test of the hypothesis implicit in Fig. 1, since as will be shown, the requirements discussed above are probably fulfilled as near as possible for a collection from twelve bodies.

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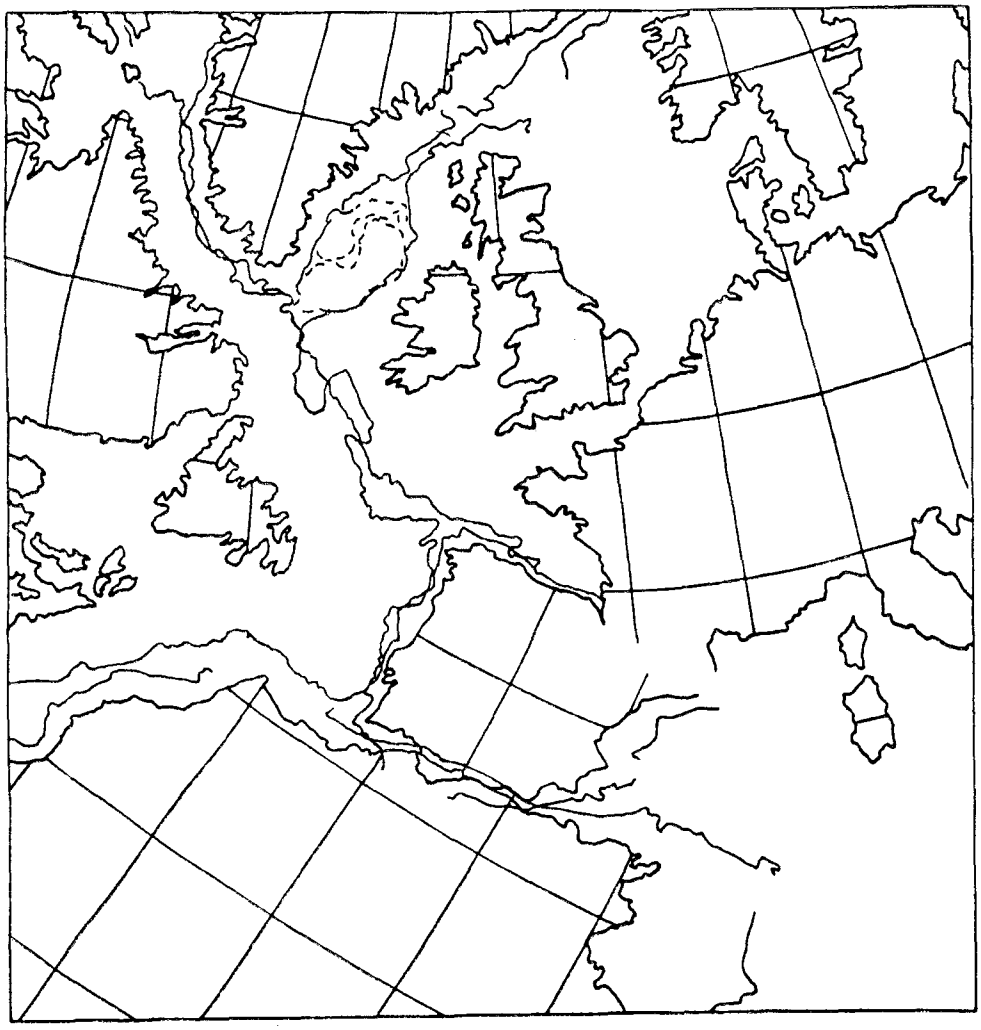


FIG. 1. Suggested pre-drift configuration of major landmasses in the North Atlantic, based on least-squares fitting of the 500 fathom contour (Bullard *et al.* 1965, by permission of the authors).

Previous work

Any palaeomagnetic data from the Iberian peninsula is pertinant to the rotation hypothesis. Girdler (1965) has summarized the limited results. Of the four surveys (Table 1) only the study of the Permo–Triassic Huesca Andesites (Schwartz 1963) and Pyrenean sediments and igneous rocks (Van der Lingen 1962) provide acceptable data. Schwartz (1963) believes that the Triassic sandstone data are unreliable, and the study of Clegg *et al.* (1957) produced a northerly NRM inclination of 57° in a region which was almost certainly in palaeomagnetically equatorial (and possibly southern) latitudes during the deposition of the sampled Triassic sandstone (Irving 1964, p. 185). As pointed out by Irving (1964, p. 256), and Girdler (1965), by comparison with results from north of the Pyrenees, the Permo-Triassic data are consistent with the rotation hypothesis, but of course can place no lower age limit on the hypothetical rotation.

In presenting Lower Tertiary isopachs and subsurface facies maps of the Basin of Aquitaine, immediately north of the Pyrenees, Schoeffler (1965) prompted com-

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Table 1

Summary of previous Iberian palaeomagnetic results

Author(s)	Site location	Rock type	Age	$D°$	$I°$	k	$\phi'°$	$\theta'°$	Remarks
Clegg *et al.* (1957)	5·0° W, 43·0° N	Sandstone	Triassic	353	+57	—	131W	82N	Data from 4 of 7 sites rejected. Stability inferred by divergence from present magnetic field direction.
Van der Lingen (1960)	0·5° W, 42·7° N	Sediments and igneous rocks (' Pyrenean ')	Permian	159	−14	—	154W	52N	
Schwartz (1963)	1·0° W, 43·1° N	Andesites (Huesca Province)	Permo-triassic	152	−22	65	133W	51N	14 specimens. Demagnetized at 900 oersteds.
Schwartz (1963)	1·0° W, 43·1° N	Red sediments (Huesca Province)	Permo-triassic	250	+51	7	54W	8N	9 specimens. Interpreted to have acquired strong post-Alpine stable NRM, obliterating the original NRM.

D, I are declination and inclination of NRM, east of north and with respect to horizontal, respectively (+ is below, and − above the horizontal). $k = N-1/N-R$ [Fisher (1953) precision parameter, where N = number of units or specimens, and R = resultant vector applying unit vector to each unit specimen]. ϕ' and θ' are longitude and latitude of resulting palaeomagnetic pole (southern hemisphere pole transposed to northern hemisphere).

ment by Funnel (1965) that the sedimentary distribution agrees with rotation of the Iberian peninsula, which Funnel believes to have taken place during the Cretaceous. Emory (1965) believes that Post-Miocene tectonics are indicated in the Basin of Aquitaine, but details of the Cretaceous and Lower Tertiary orogeny in the Pyrenees are far from perfectly understood (Schoeffler 1965). A comprehensive survey of geological literature concerning the Basin of Aquitaine can be found in publications by Vigneaux (1965), Kingebiel *et al.* (1964) and Alvinerie & Vigneaux (1966).

Geology of the area

The Lisbon Volcanics constitute a poorly outcropping but undisturbed limited series of fine grained non-porphyritic basalts and basanites, interdigitated with flat lying tuffaceous conglomerates, agglomerates, and argyllites. The lavas are restricted stratigraphically to a zone beneath the Oligocene Benfica formation and are unconformably above Upper Cenomanian carbonates. The interbasaltic sediments contain such Eocene freshwater molluscs as *Bulimus carnaxidonsis; B. olisiponensis; Andradromus riberiroi*; and *Pupa tournoueri*, as well as vertebrates. Lava thicknesses vary from 3 to 10 metres. Fig. 2. shows the geology of the area of investigation, which is due to Zbyszewski (1963, 1964).

Experimental methods

Field methods. The great amenability of the suggested tectonic history of the Iberian peninsula to palaeomagnetic study is not complimented by a ready availability of ideal palaeomagnetic materials (basalts), outcrops of which are very rare indeed. Therefore, although a limited number of lavas are involved, the opportunity presented by the Lisbon Volcanics is unique for Portugal. The amount and nature of the interbasaltic material is strong evidence that the lavas, however limited in number, span a considerable time, thereby maximizing the probability that non-axial dipole directions will be averaged out to an axial dipole, for a collection of this size.

A total of 39 samples have been collected from twelve lavas, the locations and identification numbers of which are included in Fig. 2. Each sample, taken using a petrol-powered portable drill (Doell & Cox 1965a), was oriented with respect to geographic north to an accuracy of two degrees while still attached to the outcrop.

Laboratory methods. Each core, of 2·5 cm diameter, was sliced into a specimen of 2·3 cm length for measurement of NRM, a specimen for thin section manufacture, and a specimen for Curie point determination.

The NRM was measured using an astatic and a spinner magnetometer (Foster 1966). Magnetic susceptibility was measured using a 800 c/s transformer bridge (Collinson *et al.* 1963). Unstable components of NRM in each specimen were minimized by progressive application of four values of alternating magnetic fields (McElhinny 1966) ranging from 100 to 400 oersteds. A Reichert Zetopan-Pol reflection microscope was used to examine the oxidation state of the iron–titanium oxides in each specimen, under oil at magnifications up to X1200. Each polished thin section was classified according to a high temperature titanomagnetite oxidation index series. Full details, including photomicrographs of the phases and textures involved in the classification system are given elsewhere (Watkins & Haggerty 1967). It is sufficient to mention here that the classification system consists of six indices, ranging from dominantly homogeneous unoxidized titanomagnetite (oxidation index I) to dominantly intergrowths of rutile, pseudobrookite, and hematite (oxidation index VI), with intermediate indices dependent on the abundance, nature, and composition of oxidation induced exsolution lamellae in a titanomagnetite host. Any variation in magnetic properties depends mainly on the oxidation state of the iron–titanium oxides (Watkins & Haggerty 1967).

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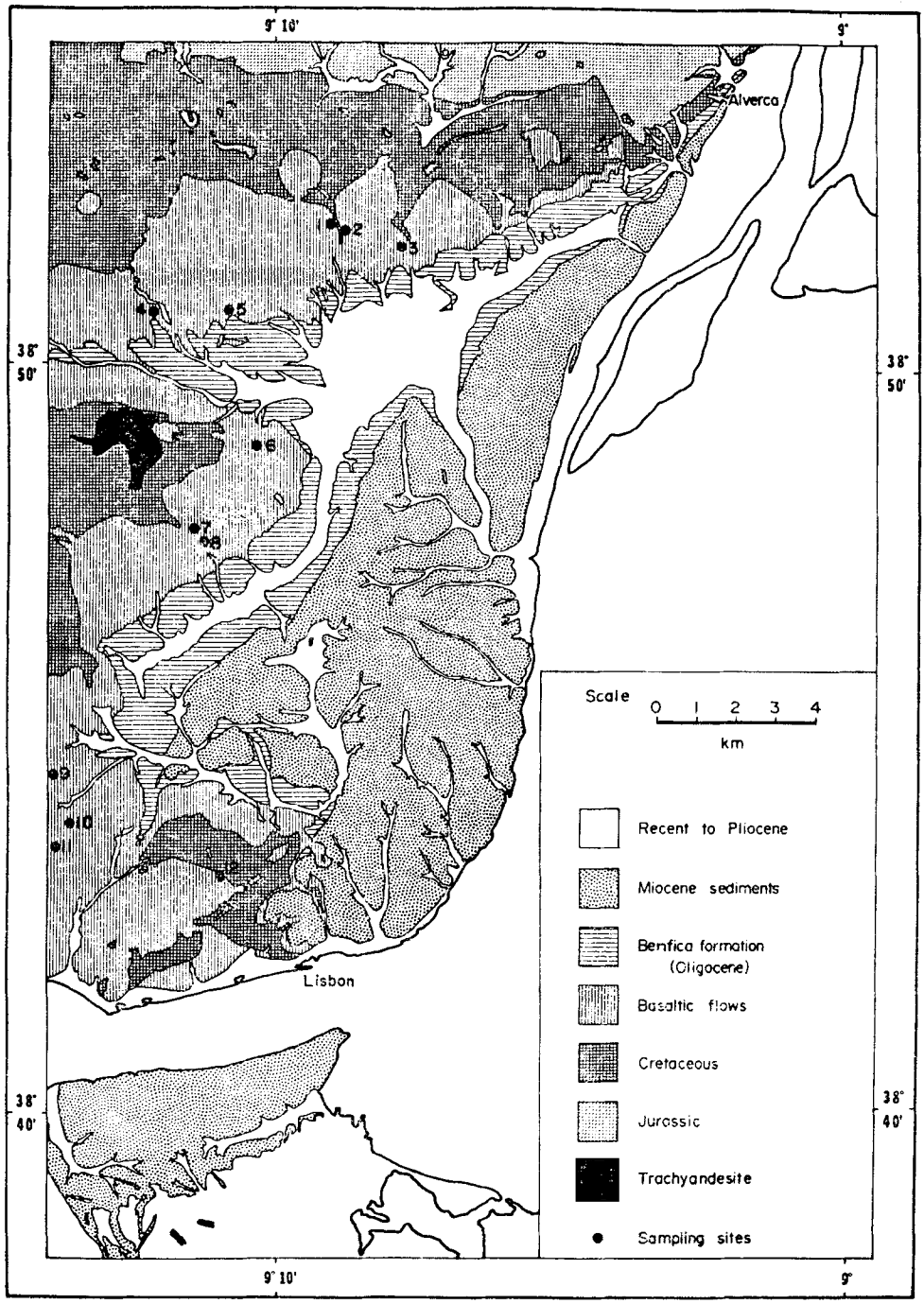


FIG. 2. Geological map of the area of investigation (Zbyszewski 1963, 1964), showing sample site locations.

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A modified Chevallier balance (Chevallier & Pierre 1932) has been used to determine the Curie points. The specimens, of approximate mass one hundred milligrams, are heated rapidly in air, to inhibit oxidizing processes. The frequent detection of low Curie points using this method (Ade-Hall *et al.* 1965) is evidence of the experimental resolution, since if an oxidation process and associated magnetic change were to take place faster than the heating rate, then only high Curie points would be detected.

All data have been reduced using a CDC 6400 computer.

Results

Fig. 3 shows the original NRM of all specimens. The values of intensity of magnetization, susceptibility, and Q factor (Koenigsberger 1938) are included in Table 2. The results of demagnetization of all specimens are given in Table 3.

Since Watkins & Haggerty (1967) have shown that a large variation in stability can exist in a single lava, and because of the fact that the higher demagnetizing fields are capable of adding spurious components of magnetization, we feel that it is more meaningful to follow essentially the suggestion of Irving *et al.* (1961) and determine the mean NRM direction in each lava using a least scatter criterion. This involves combining any one of the four NRM directions (obtained after demagnetization) with one similar NRM direction in each other specimen from the sample lava, so that the maximum resultant vector length (R) is obtained. In this way, less stable material would be expected to require a higher alternating magnetic field value to remove the viscous NRM component, to approach the true original NRM direction than would be the case with the more stable material, which incidentally may be expected to correspond to the more oxidized part of the lava (Watkins & Haggerty

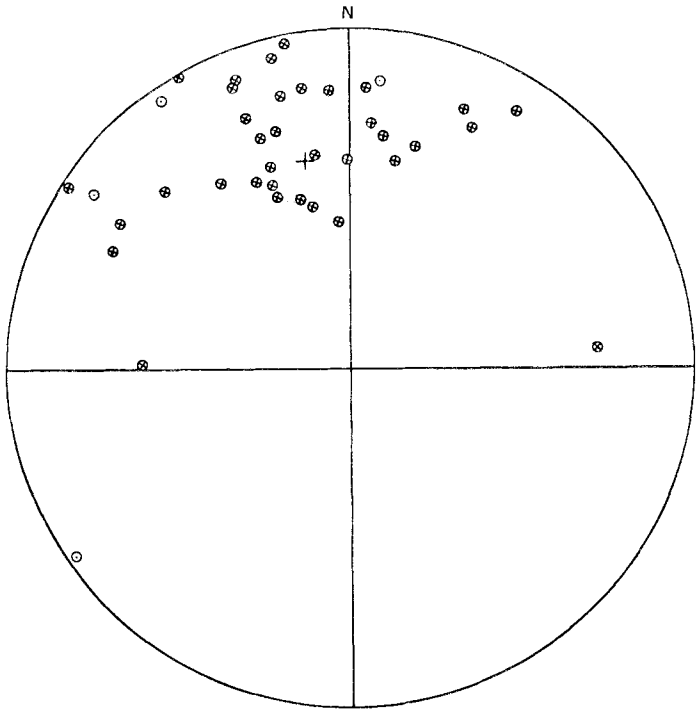


FIG. 3. Direction of original natural remanent magnetism in each specimen. ⊗ = lower hemisphere; ⊙ = upper hemisphere; + = direction of present geomagnetic field in Lisbon; N = geographic north. Projection is equal area.

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Table 2

Original natural remanent magnetism

Lava number	Sample number	$D°$	$I°$	Jo	χ	Q
1	1	337·4	44·6	2·93	1·00	6·82
	2	333·9	39·5	3·01	1·04	6·74
	3	343·4	47·6	1·28	0·74	4·03
	4	336·6	41·8	1·26	0·64	4·59
2	1	329·7	00·6	6·25	2·16	6·73
	2	2·6	18·4	1·72	4·00	1·00
	3	350·2	17·9	1·57	6·00	0·61
	4	345·6	06·8	1·18	2·64	1·05
3	1	85·6	28·7	1·99	1·50	3·09
	2	4·2	28·9	2·27	1·34	3·94
	3	7·5	31·5	2·75	1·22	5·25
4	1	315·1	35·4	1·45	1·34	2·52
	2	236·6	−03·9	1·86	1·78	2·43
	3	342·7	28·3	1·12	2·28	1·15
5	1	338·7	29·4	1·57	1·48	2·48
	2	345·9	19·4	2·32	1·50	3·60
	3	5·8	−15·3	2·90	1·62	4·17
6	1	338·2	37·6	2·43	8·60	6·59
	2	302·3	02·4	2·98	2·24	3·10
	3	355·5	54·3	0·84	9·80	2·00
7	1	337·5	22·9	4·35	1·12	9·0
	2	355·1	19·0	6·61	1·06	14·5
	3	324·2	−05·8	5·30	1·12	11·0
8	1	338·4	10·6	1·07	1·32	19·0
	2	337·6	11·5	1·59	1·40	26·4
	3	348·9	03·7	0·94	2·10	10·4
9	1	346·9	50·0	3·84	1·02	8·76
	2	297·0	23·1	2·38	1·50	3·70
	3	304·3	−09·0	4·35	1·40	7·23
10	1	358·9	38·6	4·41	1·04	9·88
	2	350·5	36·8	4·91	1·02	11·21
	3	271·1	39·2	3·19	0·98	7·57
11	1	32·5	10·1	3·32	1·76	4·39
	2	26·6	21·5	4·33	1·74	5·79
	3	11·8	37·6	2·68	1·44	4·33
	4	15·6	32·6	2·99	1·62	4·30
	5	23·1	17·9	5·68	1·72	7·68
12	1	314·1	24·9	3·26	0·21	36·20
	2	302·2	20·4	3·93	0·71	9·61

Lava number corresponds to site location shown in Fig. 2. D, I are declination and inclination of NRM, in degrees east of geographic north, and with respect to the horizontal, respectively (negative degrees are upwards). Jo = intensity of magnetization in e.m.u./g$\times 10^{-3}$. χ = mass susceptibility in millioersteds per gauss. $Q = Jo/\chi H$ where H = present geomagnetic field intensity in Lisbon (0·43 gauss).

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Table 3

Results of treatment of each specimen in alternating magnetic fields

Lava No.	Sample No.	100 oersteds			200 oersteds			300 oersteds			400 oersteds		
		$D°$	$I°$	J/Jo	$D°$	$I°$	J/Jo	$D°$	$I°$	J/Jo	$D°$	$I°$	J/Jo
1	1	338·4	45·5	0·92	338·8	45·2	0·69	345·8	49·8	0·12	354·7	55·0	0·05
	2	340·3	41·6	0·90	341·4	43·2	0·70	9·2	48·0	0·12	349·0	52·8	0·05
	3	351·2	46·8	1·08	348·3	47·7	0·87	350·4	49·0	0·14	349·7	48·6	0·16
	4	344·8	43·7	1·13	346·8	41·8	0·98	352·2	41·9	0·21	352·9	44·3	0·29
2	1	345·5	21·7	0·16	340·7	15·4	0·06	298·5	14·1	0·02	202·9	02·0	0·01
	2	353·0	41·8	0·39	231·4	−06·6	0·26	350·6	30·1	0·08	322·5	53·9	0·07
	3	344·6	36·3	0·46	345·4	28·0	0·25	344·3	55·6	0·08	332·7	39·6	0·04
	4	353·4	29·6	0·51	355·0	37·6	0·23	9·1	50·3	0·08	51·0	43·8	0·07
3	1	4·3	56·6	0·35	357·2	59·9	0·21	26·5	73·1	0·02	219·7	48·3	0·07
	2	355·9	44·1	0·31	2·9	51·6	0·17	6·2	55·0	0·03	139·1	60·9	0·07
	3	9·5	54·5	0·17	30·3	62·9	0·07	323·3	0·34	0·03	167·6	−22·0	0·05
4	1	325·4	33·9	0·36	325·6	33·6	0·23	293·5	44·8	0·08	332·1	12·3	0·10
	2	320·9	31·2	0·25	316·2	36·5	0·15	281·6	38·6	0·05	310·1	33·4	0·05
	3	330·2	29·9	0·39	313·6	26·0	0·19	291·8	46·1	0·06	308·3	−08·0	0·04
5	1	351·3	14·1	0·87	351·8	13·8	0·64	349·9	12·7	0·21	352·1	17·6	0·13
	2	353·6	27·2	0·20	350·5	26·4	0·12	342·7	09·7	0·04	357·1	09·5	0·05
	3	357·8	00·3	0·46	314·5	02·1	0·28	170·4	04·7	0·07	176·7	−00·3	0·06
6	1	343·8	40·3	0·93	343·5	41·6	0·72	344·4	39·4	0·26	343·9	40·8	0·21
	2	348·8	36·0	0·56	349·8	42·1	0·33	350·4	43·1	0·10	356·7	44·0	0·10
	3	356·3	50·0	0·45	357·7	52·0	0·28	11·2	61·5	0·07	9·6	49·1	0·08
7	1	342·1	26·7	0·12	344·8	07·7	0·02	305·2	43·8	0·01	29·1	27·8	0·01
	2	356·3	23·6	0·10	4·8	19·4	0·07	258·7	71·9	0·01	291·4	14·7	0·01
	3	327·9	13·1	0·09	306·0	23·1	0·04	258·3	29·2	0·02	323·5	46·8	0·02
8	1	336·2	11·4	0·89	336·0	10·0	0·66	333·7	09·5	0·17	333·2	13·5	0·10
	2	337·8	13·1	0·95	337·0	10·9	0·86	334·4	11·2	0·77	334·2	11·6	0·48
	3	347·5	05·8	0·88	347·0	01·8	0·59	344·7	03·7	0·20	344·3	01·4	0·01
9	1	356·2	50·8	0·09	9·0	52·2	0·05	43·9	56·2	0·04	55·3	41·3	0·02
	2	350·1	48·1	0·13	7·0	55·8	0·11	8·2	58·4	0·09	37·5	50·3	0·03
	3	343·5	37·0	0·07	356·6	38·8	0·05	12·8	27·7	0·04	358·9	−09·1	0·01
10	1	358·5	36·6	0·33	5·7	35·9	0·10	324·9	75·1	0·05	38·5	25·4	0·02
	2	351·1	38·0	0·29	348·2	32·6	0·06	347·5	13·5	0·02	323·6	−18·2	0·01
	3	277·2	41·7	0·21	286·2	44·0	0·07	164·1	51·9	0·02	273·6	38·8	0·03
11	1	29·1	50·0	0·23	22·1	55·1	0·14	80·4	76·5	0·08	153·9	07·2	0·03
	2	21·4	47·1	0·53	23·8	50·1	0·23	39·4	58·7	0·03	22·7	32·1	0·02
	3	6·6	47·5	0·50	14·0	47·3	0·21	12·6	55·1	0·05	358·3	42·8	0·03
	4	18·7	50·5	0·57	17·4	55·2	0·26	27·2	58·2	0·06	23·4	64·3	0·02
	5	7·7	39·2	0·43	4·8	45·2	0·22	3·0	54·8	0·03	345·9	28·7	0·02
12	1	345·7	33·9	0·39	347·0	35·9	0·18	350·8	−71·6	0·04	332·2	32·9	0·03
	2	347·4	39·0	0·16	286·7	57·8	0·08	278·2	26·8	0·05	298·1	40·0	0·03

D and I = declination and inclination of NRM, east of geographic north and with respect to the horizontal (negative indicates above the horizontal; otherwise, below the horizontal). J = intensity of magnetization following application of the given magnetic field (100 to 400 oesteds). Jo = intensity of magnetization in untreated specimen.

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1967). The original (untreated) NRM cannot be included in this computation since it is conceivable that a large viscous component could create a mean NRM which is not in the original ambient geomagnetic field direction, but which is also of very close grouping. Directions of the least scatter NRM are presented in Fig. 4.

The oxidation index for each specimen is presented in Table 4, together with the estimated titanomagnetite (or derivative) grain size average, and the limited Curie point data.

Discussion

The lavas are of normal polarity only.

The demagnetization data, presented in Table 3, show the presence of large unstable components, since large fractions of the original intensity of magnetization (*Jo*) are removed by the demagnetizing process. There is no obvious correlation between intensity of magnetization (*Jo*) and stability, as observed in some other lavas (Watkins & Haggerty 1967). On close examination, however, a definite correlation emerges between the character of the coercivity spectra and *Jo*. This point is explained and illustrated in Fig. 5. Negative ΔJ values (as defined in Fig. 5) are strongly associated with the lower *Jo* values, whereas those specimens with *Jo* values in excess of $3{\cdot}5 \times 10^{-3}$ e.m.u./g are, with only one exception associated with positive ΔJ values only. This shows that only in those specimens with low *Jo* does a larger fraction of *Jo* reside in the higher coercivity components. This can be interpreted to be due to the higher *Jo* values being largely a function of unstable, low coercivity (viscous) components acting in approximately the same normal polarity direction as the originally acquired NRM, thereby adding to the original *Jo*, and producing the steep gradient in the J/Jo: H curves, between $H = 100$ and 200 oersteds. The sign of ΔJ for each specimen is included in Table 4.

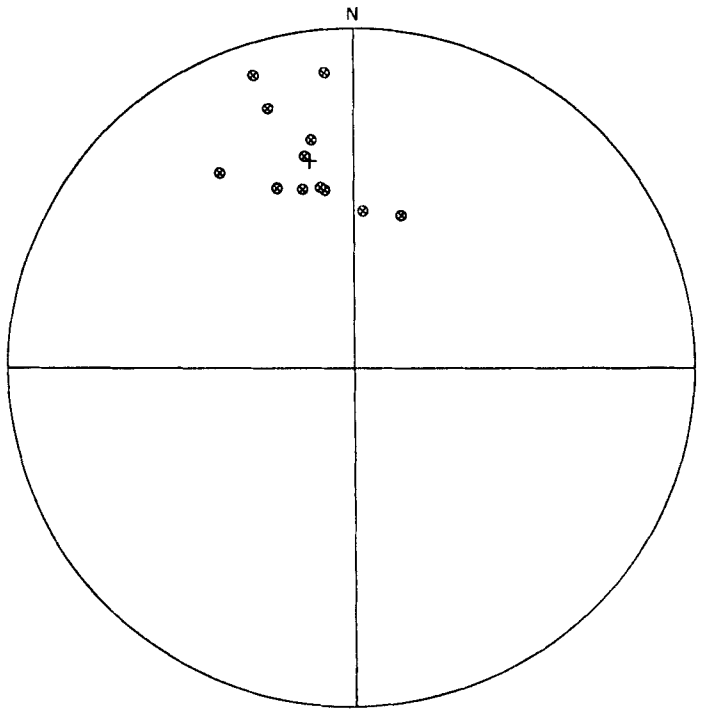


FIG. 4. Directions of mean NRM in each lava. The directions are computed using a ' least scatter ' criteria (see text). Symbols as in Fig. 3.

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Table 4

Oxidation index, estimated titanomagnetite (or derivative) grain size average, Curie point and type, and coercivity spectra sign

Lava number	Specimen number	Oxidation index	Titano grain size (μ)	Curie point (s) (°C)	Curie point type	Coercivity spectra sign
1	1	II	80			−
	2	I	70			−
	3	IV	75			−
	4	III	50	625	(1a)	−
2	1	M	20			+
	2	I	20	650	(2a)	−
	3	II	25			+
	4	M	30			+
3	1	I	30	250 550	(3a)	−
	2	I	20			None
	3	I	30			+
4	1	I	30			−
	2	II	30	400 600	(3b)	None
	3	I	10			+
5	1	M	30			−
	2	IV	40	570	(1a)	None
	3	II	30			−
6	1	III	22			−
	2	IV	30			None
	3	(?)	100	570	(1b)	−
7	1	I	60			+
	2	I	100	420	(1b)	−
	3	II	80			+
8	1	I	150			−
	2	IV	50			None
	3	II	80	610	(1b)	−
9	1	II	30			+
	2	I	25			−
	3	II	30	450	(2a)	+
10	1	I	20	570	(1b)	+
	2	I	15			+
	3	I	20			+
11	1	I	40			+
	2	I	80			+
	3	I	50			+
	4	I	40			+
	5	II	50	400 650	(3b)	+
12	1	M	100	375	(1b)	+
	2	I	25			+

For explanation of oxidation index see text; M denotes dominance of maghemite, rendering high temperature titanomagnetite oxidation index indeterminable; (?) denotes unclassifiable because of great range of oxidation indices within the given polished thin section. For explanation of Curie point types see text. For explanation of coercivity spectra sign see Fig. 5. 'None' indicates $A = B$ (Fig. 5).

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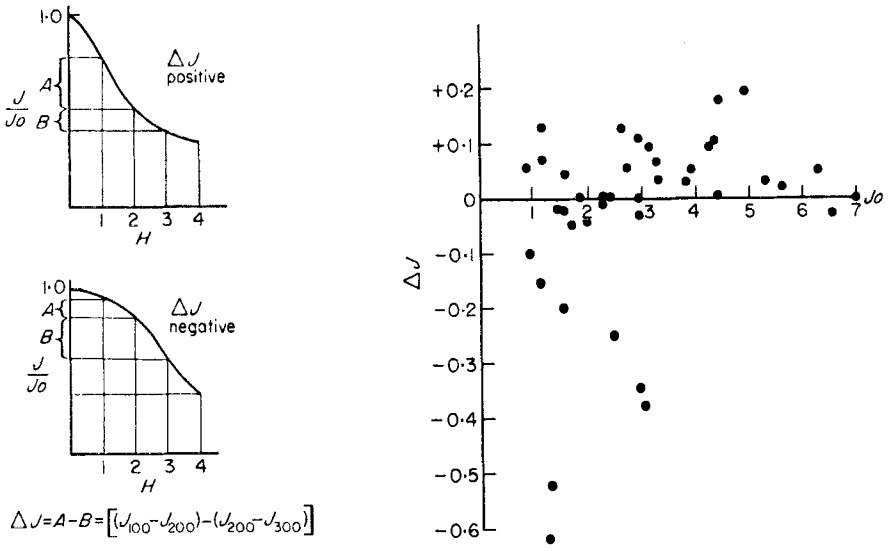


FIG. 5. Variation of the character of the demagnetization curves and their correlation with the intensity of magnetization (J_0). H = demagnetizing field in hundreds of oersteds. For further explanation, see text.

The iron–titanium oxides are mostly in the form of euhedral titanomagnetite grains, commonly of average size greater than 25μ, with low oxidation states dominating, if not being present throughout (Table 4). Titanomaghemite, when present, appears to occur in conjunction with the lower oxidation state titanomagnetites only. In a few cases (Table 4) titanomaghemite is so well developed that no high temperature oxidation index can be assigned. Most of the low oxidation state specimens are magnetically unstable, as can be seen from comparison of Table 3 and 4. Large fractions of *Jo* are destroyed by application of alternating magnetic fields when a low oxidation state exists; but the higher oxidation state specimens almost always retain large fractions of *Jo* upon similar treatment. The correlation of low oxidation state and low stability is to be expected, according to the results obtained by Watkins & Haggerty (1967), and Larson *et al.* (1966), who have shown that increasing oxidation in basalts, resulting in a diminished effective grain size through the subdivision of titanomagnetite by ilmenite lamellae, is associated with increased stability.

In addition to presenting Curie points for specimens from each lava in Table 4, the Curie point curves (four examples of which are shown in Fig. 6) have been classified essentially according to the convention proposed by Doell & Cox (1965b): Type 1 is a simple curve with a well-defined knee, consistent with a single ferromagnetic constituent; type 2 possesses no well-defined knee, being consistent with a variable ferromagnetic composition; type 3 possesses a double knee curve, being consistent with two ferromagnetic components. Type 3 can conceivably be also due to a low Curie point constituent which increases to a higher value by oxidation during the heating process. Because of the fact that the apparatus does not employ a vacuum system, the cooling part of the cycle may be expected to show the effect of any oxidation caused by the heating. If the thermal remanent magnetism (TRM) on cooling is less than the original NRM then the suffix (a) is added to the type number; and the suffix (b) is added when the TRM on cooling is greater than the original NRM. The exact meaning of such changes, if any, is conjectural, but would appear to be most likely an expression of the sign and magnitude of the phase produced or destroyed by the heating process. One would expect only the highest oxidation state titanomagnetites to yield identical heating and cooling curves.

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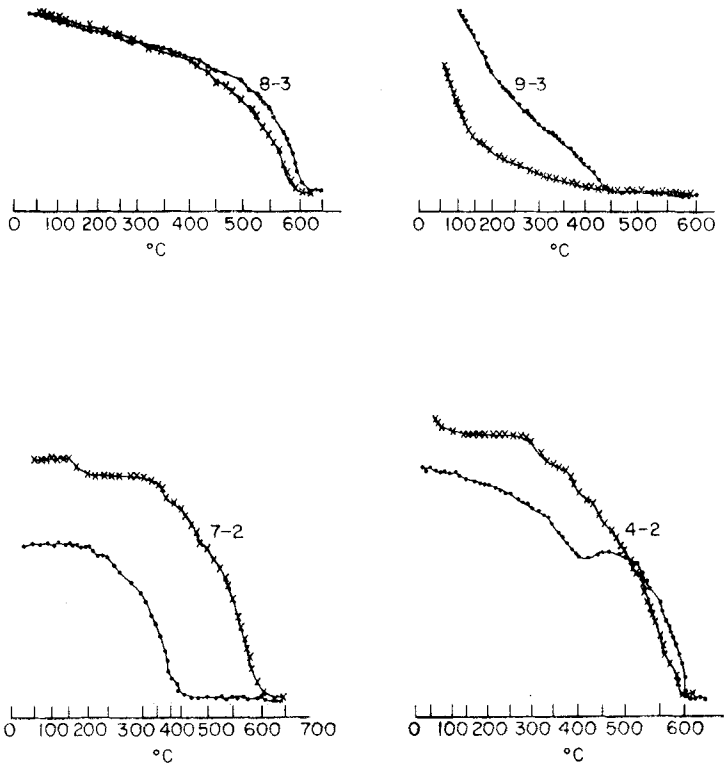


FIG. 6. Four tracings of original Curie point data: heating curve defined by dots, cooling curve defined by crosses. Absissae in degrees centigrade; ordinates in arbitrary units which are a function of intensity of magnetization. Using the classification described in the text, the curves are defined as follows: 8–3 (1b), 9–3 (2a), 7–2 (1b), 4–2 (3b). The specimen numbering convention is: first number is the lava (Fig. 2); second number is the core.

The fact that widely differing Curie points, widely differing stabilities, and a range of oxidation states occur in lavas of the same polarity is strong evidence that the polarity is independent of the ferromagnetic constituents, and confirms the minimum scatter (K_{max}) computational procedure as being more realistic than applying a single demagnetizing field to all specimens, since the latter process would include an assumption of uniformity of coercivity. There is no reason to suspect that the NRM is other than that due to the original Eocene geomagnetic field. A long period of normal polarity is suggested for part of the Eocene period.

Table 5 shows the results of the application of Fisher (1953) statistical analyses to the data. The palaeomagnetic pole and its associated 95 per cent oval of confidence is included in Fig. 7 which also shows the virtual geomagnetic poles for each lava, and the mean Lower Tertiary palaeomagnetic pole position for European rocks, according to Irving (1964, p. 185). The latitude of the mean palaeomagnetic pole for the Lisbon volcanic agrees with the other European results, but the longitude is clearly at variance with the independent data. The declination result is well defined because of the low inclination values. The cause of such a disagreement in local and regional NRM declinations can be explained by one of four possibilities:

1. The lavas sampled do not span sufficient time to average out non-axial components, and the observed mean NRM direction merely represents an Eocene geomagnetic field component other than the axial dipole. Within the 95 per cent

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Table 5(a)

Statistical analyses (each lava)

Lava number	K_0	K_{100}	K_{200}	K_{300}	K_{400}	K_{max}	D	I	ϕ'	θ'	θ_{63}
1	328·5	309·9	408·2	110·1	260·7	528·9	343·3	45·0	224·7	71·4	3·52
2	26·1	72·3	2·3	7·1	1·8	227·6	348·7	27·6	196·4	63·9	5·37
3	4·4	103·0	58·7	3·9	2·4	1909·8	6·7	55·4	107·1	83·9	1·85
4	2·3	328·2	113·1	169·3	11·2	338·9	325·6	31·6	234·7	53·0	4·40
5	9·1	34·5	11·6	1·0	1·0	113·6	355·6	7·8	178·5	54·9	7·60
6	5·6	91·6	112·6	30·9	66·0	163·2	349·8	45·0	208·9	75·1	6·34
7	14·1	28·9	7·8	8·4	3·2	28·9	341·8	21·5	206·1	58·0	15·07
8	114·8	128·8	109·1	124·5	82·9	174·4	339·3	8·2	204·6	50·5	6·13
9	4·8	89·3	65·3	15·1	4·0	143·0	354·4	45·9	194·7	77·6	6·77
10	5·1	6·0	6·6	2·2	1·8	7·8	321·1	56·1	269·2	59·4	29·07
11	36·6	106·0	147·5	30·5	2·5	226·0	16·0	54·3	93·1	76·6	5·39
12	93·4	471·1	6·6	1·2	16·6	1352·3	347·2	37·5	206·1	69·0	2·20

Lava number corresponds to site location shown in Fig. 1. K = Fisher (1953) precision parameter = $N-1/N-R$ where N = number of specimens, R = resultant vector, applying unit vector each specimen. Suffixes 0, 100, 200, 300, 400 indicate demagnetizing field values applied to each specimen; and max indicates the precision parameter obtained using a ' least scatter ' criterion (see text). D and I are declination and inclination of ' least scatter ' NRM, and ϕ' and θ' are the corresponding palaeolongitude and palaeolatitude of the virtual geomagnetic pole. θ_{63} is the circular standard deviation.

Table 5(b)

Statistical analysis (all lavas combined)

Treatment	D	I	R	K	ω	β	α_{95}	d_p	d_m	ϕ'	θ'
0	338·1	27·3	10·78	8·99	8·68	15·12	15·3	9·1	16·7	215·2	58·9
100	346·8	35·9	11·52	22·9	42·4	26·6	9·3	6·2	10·8	205·6	67·9
200	341·6	38·1	11·26	14·8	14·85	26·2	11·6	8·1	13·8	218·4	66·5
300	334·6	44·8	10·05	5·6	5·14	17·0	20·1	16·0	25·3	238·1	65·4
400	341·7	43·5	9·78	4·95	3·14	20·2	21·8	16·9	27·1	225·0	69·6
Combined	346·7	37·2	11·31	15·95	63·33	17·09	11·2	7·7	13·2	207·0	68·7

Treatment refers to alternating magnetic field value applied to each specimen, except ' combined ' which refers to the combined twelve lavas, using the ' least ' scatter NRM direction values for each lava. D and I are declination and inclination values. R = resultant vector, applying unit vector to each lava. $K = N-1/N-R$ where N = number of specimens. ω and β = within and between lava confidence parameters (Watson & Irving, 1957). α_{95} = semi-vertical angle of the 95 per cent confidence oval, parallel and normal to D, around the palaeomagnetic pole, which is at longitude ϕ', latitude θ'.

confidence oval, this possibility is not considered to be a likely explanation, since as indicated earlier, the interbasaltic sediments provide strong evidence that the section represents a considerable time span.

2. The original NRM has been modified by Post-Eocene processes, and the observed mean NRM represents a magnetic field younger than Eocene. Tectonic disturbances locally are minimal, however, and no Post-Eocene intrusion or other thermal sources are present locally. The presence of low-oxidation state titanomagnetites is also evidence pointing to a lack of thermal changes of the original NRM.

3. The geomagnetic field during the extrusion of the lavas was significantly west of the Eocene geomagnetic field which has been recorded by other European rocks (see Irving 1964, p. 310).

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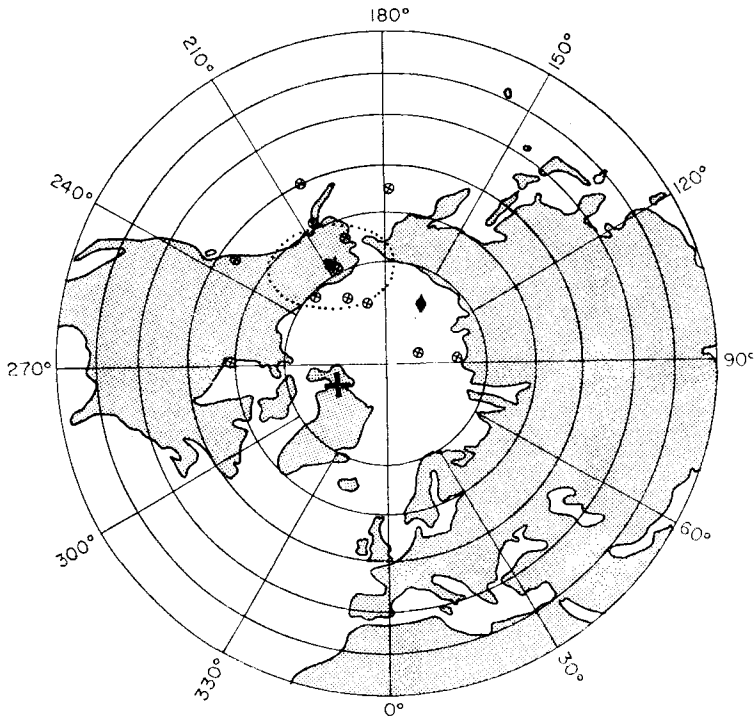


FIG. 7. Virtual geomagnetic poles for each lava, following demagnetization of each specimen in 100, 200, 300, and 400 oersteds, and least scatter direction computation (see text). Mean of all virtual geomagnetic poles (the palaeomagnetic pole) shown as a star, together with the associated 95 per cent confidence oval. The mean Lower Tertiary palaeomagnetic pole for Europe (Irving 1964, p. 185) shown as the solid diamond. Thick cross is position of the present geomagnetic pole. Projection is equal area.

4. The mean NRM represents an Eocene geomagnetic field which approximates an axial dipole configuration, but the area has experienced rotation about a vertical axis, to result in a divergence between the mean NRM declination in the Lisbon Volcanics, and that determined from the other European Eocene collections. This possibility is examined in Fig. 8.

In Fig. 8 the mean Lower Tertiary palaeomagnetic pole position for Europe, according to Irving (1964 p. 185) has been used to construct the Lower Tertiary isoclines, or lines of equal magnetic inclination, on the assumption that the Lower Tertiary palaeomagnetic pole is due to a geocentric dipole. Arbitrarily spaced magnetic meridians have been added, as well as the sites of most of the Lower Tertiary rock collections, used in defining the position of mean associated palaeomagnetic pole position. The NRM declination resulting from the present study is shown at the sample site, together with the declinations corresponding to the 95 per cent confidence oval limits. In order to bring the mean NRM declination into coincidence with the Lower Tertiary magnetic meridians, which are 9° east of north, the Iberian peninsula has been rotated by about 22°. This also results in a closer agreement between the mean inclination of NRM and the Eocene isocline value, than would be the case in the unrotated situation.

This fourth possibility is strongly consistent with the rotation hypothesis of Du Toit (1938), Carey (1955), and Bullard *et al.* (1965), and would imply that only about half of the suggested rotation was completed by Eocene times. The configuration

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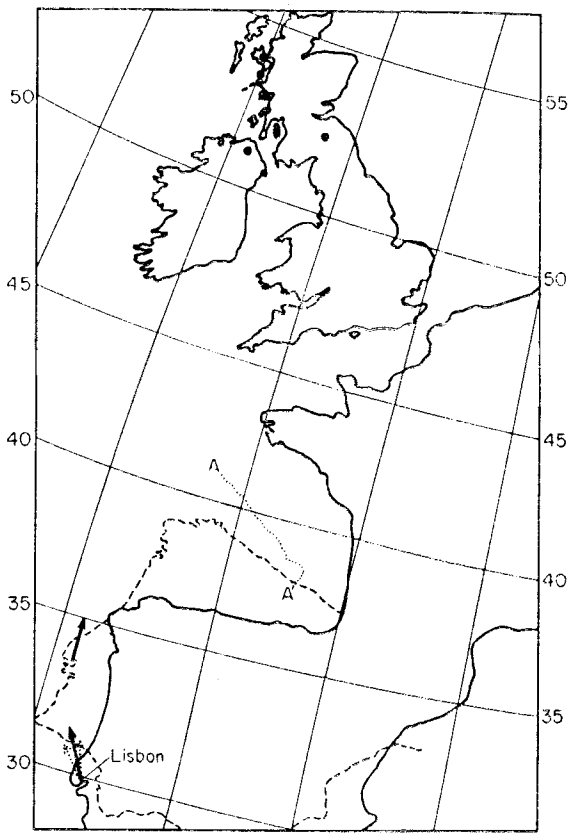


FIG. 8. Sketch illustrating a possible explanation of the observed palaeomagnetic declination of the Lisbon Volcanics. The mean Lower Tertiary palaeomagnetic pole (Irving 1964, p. 185) has been used to construct corresponding isoclinical lines (lines of equal magnetic inclination) at 5° intervals and arbitrary meridians. The observed palaeomagnetic declination for the Lisbon Volcanics is shown at the present Lisbon position, and includes the 95 per cent confidence limits. The observed declination has been rotated, with the Iberian peninsula (shown by dotted line) into parallelism with the Lower Tertiary meridians. A–A′ is the present 2000 ft bathymetric contour. The sample sites within the area illustrated which were used in defining the Lower Tertiary palaeomagnetic pole position are shown as solid circles.

of the present 2000 ft submarine contour (AA′), shown in Fig. 8, would suggest active opening of the Biscay Sphenochasm (Carey 1958) at that time.

When totally independent approaches provide the same conclusion, the hypothesis involved becomes increasingly difficult to reject. Such a rotation can be envisaged as the result of differential crustal spreading (Pitman & Heirtzler 1966, Fig. 5(b)) which, assuming a uniform rotation rate since the Eocene, would amount to rotation of less than one half of a degree per million years with a maximum linear spreading rate of the order of 1 cm per year for northwestern Spain.

The deliberate choice of the term ' consistent with ' (above) is emphasized and should not be interpreted to mean that the authors believe the data constitute incontravertible evidence of the validity of the rotation hypothesis. Quite obviously the complexities of major hypotheses demand several approaches all of which must be considered in the final assessment of the merits of the hypothesis, and generally no single method will alone provide final proof of the validity or failure of a hypothesis. The necessity of invoking a hitherto unused or unrealized tectonic process

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should never, however, be entertained as an argument against a new hypothesis, particularly if supported by even a limited set of data. To do so is to assume a perfect understanding of natural processes.

Conclusions

As a result of this study the following conclusions are made:

1. During the Eocene period in Portugal an alternating sequence of lavas, fossiferous sediments, and pyroclastics were formed at a time of normal geomagnetic polarity.

2. The palaeomagnetic pole obtained from the lavas can be explained either by anomalous Eocene geomagnetic field behaviour or by a 20° or so anticlockwise rotation of the Iberian peninsula since the extrusion of the lavas. This latter explanation is about one half of the rotation independently suggested by Du Toit (1938), Carey (1955), and Bullard *et al.* (1965), on totally independent grounds; and can therefore be interpreted as supporting a rotation of the Iberian peninsula which was initiated prior to the Eocene, and which continued after that time.

The geology of the Basin of Aquitaine, although not in conflict with this possibility is still insufficiently well defined to assist in further resolution of the hypothesis, and consequently may be profitably investigated, (as is also the case in the complex Pyrenean geology) with post-Eocene rotation in mind.

Aeromagnetic or marine magnetic coverage of the Bay of Biscay may provide pertinent data, since if differential crustal spreading has occurred south of the 2000 ft submarine countour (AA′ in Fig. 5), a record of the Earth's changing geomagnetic polarity may be preserved in such a spreading crust in the form of linear magnetic anomalies, as may have happened in other active areas (Vine, 1966). These would be slightly inclined to being fan-shaped and with a long axis from northwest to southeast. It is conceivable that Pre- and Post-Eocene geomagnetic polarity changes, the sequence of which is thought to be recorded elswhere (Heirtzler, 1967), may be preserved in the oceanic crust of the Bay of Biscay.

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Note added in proof

Since the original manuscript was submitted for publication, Mr E. J. W. Jones (Department of Geodesy and Geophysics, University of Cambridge), has generously made available to us a description of some highly relevant unpublished data (Jones & Funnell 1968). In a letter of 1967 November 11, Jones informs us that Upper Cretaceous sediments exist on a seamount (which he calls the 'Cantabria Seamount'), at latitude 45° 04·5′ North, longitude 8° 00·1′ West. He also describes the existance of a seismically defined horizon of suspected Upper Cretaceous age, beneath the 'Gascony Seamount' at approximately 46° 35′ North, 5° 20′ West. If it can be shown that a Cretaceous horizon underlies the entire Bay of Biscay and adjacent French continental shelf, then the possibility of Post-Eocene rotation of the Iberian peninsula

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would appear to be precluded, and the palaeomagnetism of the Lisbon Volcanics would therefore reflect an unusual Eocene geomagnetic field behaviour. On the other hand if the Upper Cretaceous occurrences, as detailed above, actually flank a central northwest–southeast trending fan-shaped zone of progressively younger crustal material, then the possibility of Post-Eocene rotation of the Iberian peninsula, with a corresponding creation of oceanic crust, would be supported. Similarly, if the age of the oceanic crust becomes progressively younger northeastwards from the Gascony Seamount towards the continental shelf and French coast, differential crustal spreading and Post-Eocene rotation of the Iberian peninsula might be suspected. These ambiguities are clearly capable of being resolved by additional appropriate data. Mr Jones is acknowledged for his valuable comments.

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Jones, E. J. W. & Funnell, B. M., 1968. Association of a seismic reflector and Upper Cretaceous sediment in the Bay of Biscay, submitted for publication to *Deep Sea Res.*

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